$$E_B = a_V A - a_S A^{2/3} - a_C \frac{Z(Z-1)}{A^{1/3}} - a_A \frac{(N-Z)^2}{A}$$

$$\mathcal{E} = n_N M(A,Z) + \mathcal{E}_e(n_e) + \mathcal{E}_n(n_n) + \mathcal{E}_c(n_e)$$

$$E_c = -\frac{9}{10}\frac{Ze^2}{r_0} \qquad \frac{E_c}{V} = -\frac{9}{10}\frac{Ze^2}{r_0}\frac{3}{4\pi r_0^3} = -\frac{27Ze^2}{40\pi r_0^4}$$

$$r_0 = \sqrt[3]{\frac{3Z}{4\pi n_e}} = \left(\frac{3Z}{4\pi}\right)^{1/3} n_e^{-1/3}$$

$$\frac{E_c}{V} = -\frac{27Ze^2}{40\pi}\left(\frac{4\pi}{3Z}\right)^{4/3} n_e^{4/3} \qquad \mathcal{E}_c = -\frac{27}{40}\frac{kZe^2}{r}\left(\frac{4\pi}{3Z}\right)^{4/3} n_e^{4/3}$$

$$n = n_N A + n_n \qquad n_e = n(1-Y_n)\frac{Z}{A} \qquad Y_n = \frac{n_n}{n}$$

$$n_N = \frac{n_e}{Z} \qquad n_e = Z n_N$$

$$\frac{4}{3} - 1 = \frac{1}{3}$$

$\mathcal{E}(n, A, Z, Y_n)$ constant n

$$\frac{\partial \mathcal{E}}{\partial Z} = \frac{n(1-Y_n)}{A}\frac{\partial M(A,Z)}{\partial Z} + \frac{\partial \mathcal{E}(n_e)}{\partial n_e}\frac{\partial n_e}{\partial Z} + \underbrace{\frac{\partial \mathcal{E}(n_n)}{\partial Z}}_{0}$$

$$+ -\frac{27}{40}\frac{kZe^2}{r}\left(\frac{4\pi}{3Z}\right)^{4/3}\frac{\partial n_e^{4/3}}{\partial n_e}\frac{\partial n_e}{\partial Z} = 0$$

$$\frac{\partial n_e}{\partial Z} = \frac{n(1-Y_n)}{A}$$

$$\frac{\partial M(A,Z)}{\partial Z} + \epsilon_{F,e} - m_e c^2 + -\frac{27}{40}\frac{kZe^2}{r}\left(\frac{4\pi}{3Z}\right)^{4/3}\left(\frac{4}{3}r_e^{1/3}\right)$$

$$= \frac{\partial M(A,Z)}{\partial Z} + \epsilon_{F,e} - m_e c^2 - \frac{9}{10}\frac{kZe^2}{r}\left(\frac{4\pi}{3Z}\right)^{4/3} r_e^{1/3} = 0$$

$$n_e = 2\left(\frac{4\pi}{3}\right)\left(\frac{p_{F,e}}{h}\right)^3 \qquad x = \frac{p_F}{m_e c} \qquad p_F = m_e c x$$

$$n_e = 2\left(\frac{4\pi}{3}\right)\left(\frac{m_e c}{h}\right)^3 x^3 \qquad \epsilon_{F,e} = m_e c^2\sqrt{1+x^2}$$

$$Z^{2-1/3} \qquad Z = \frac{4}{3} = \frac{6-4}{3}$$

$$\frac{\partial \varepsilon}{\partial A} = n(1-Y_n)\frac{\partial}{\partial A}\left(\frac{M(A,Z)}{A}\right) + \frac{\partial \bar{\varepsilon}(n_e)}{\partial n_e}\frac{\partial n_e}{\partial A} + \underbrace{\frac{\partial \bar{\varepsilon}(n_n)}{\partial n_n}\frac{\partial n_n}{\partial A}}_{0}$$

$$+ \frac{-2\pi}{40}\frac{\hbar^2 Z^2 e^2}{r}\left(\frac{4\pi}{3Z}\right)^{1/3} n_e^{1/3}\frac{\partial n_e}{\partial n_e}\frac{\partial n_e}{\partial A} = 0 \qquad n_e = \frac{n(1-Y_n)Z}{A}$$

$$\frac{\partial n_e}{\partial A} = -\frac{n(1-Y_n)Z}{A^2} \qquad Y_n = \frac{n_n}{n}$$

$$\frac{\partial}{\partial A}\left(\frac{M(A,Z)}{A}\right) = \frac{1}{A^2}\left(A\frac{\partial M(A,Z)}{\partial A} - M(A,Z)\frac{\partial A}{\partial A}\right)$$

$$= \frac{1}{A}\frac{\partial M(A,Z)}{\partial A} - \frac{M(A,Z)}{A^2} \qquad n_n = nY_n$$

$$\frac{1}{A}\frac{\partial M(A,Z)}{\partial A} - \frac{M(A,Z)}{A^2} + (\varepsilon_{F,e} - m_e c^2)\left(-\frac{Z}{A^2}\right)$$

$$-\frac{9}{10}\frac{\hbar^2 Z^2 e^2}{r}\left(\frac{4\pi}{3Z}\right)^{1/3} n_e^{1/3}\left(-\frac{Z}{A^2}\right) = 0$$

$$n_e = \frac{8\pi}{3}\left(\frac{m_e c}{h}\right)^3 x^3$$

$$A\frac{\partial M(A,Z)}{\partial A} - M(A,Z) - Z(\varepsilon_{F,e} - m_e c^2)$$

$$+\frac{9}{10}\frac{\hbar^2 Z^2 e^2}{r_e}\left(\frac{4\pi}{3Z}\right)^{1/3} Z n_e^{1/3} = 0 \qquad 2 - \frac{4}{3} = \frac{6-4}{3} = \frac{2}{3}$$

$$\frac{2}{3} + 1 = \frac{2+3}{3}$$

$$\frac{\partial n_e}{\partial Y_n} = \frac{\partial}{\partial Y_n}\left(\frac{nZ}{A} - \frac{nZ}{A}Y_n\right) = -\frac{nZ}{A} \qquad \frac{\partial n_n}{\partial Y_n} = n^{-\frac{5}{3}}$$

$$\frac{\partial \varepsilon}{\partial Y_n} = -\frac{n}{A}M(A,Z) + \frac{\partial \bar{\varepsilon}(n_e)}{\partial n_e}\left(-\frac{nZ}{A}\right) + \frac{\partial \bar{\varepsilon}(n_n)}{\partial n_n}n$$

$$-\frac{9}{10}\frac{\hbar^2 Z^2 e^2}{r}\left(\frac{4\pi}{3Z}\right)^{1/3} n_e^{1/3}\left(-\frac{nZ}{A}\right) = 0$$

$$\frac{m(A,Z)}{A} = m_n c^2 - a_v + \frac{Z}{A}(m_p + m_e - m_n)c^2 + a_s A^{-1/3} + a_c \frac{Z(Z-1)}{A^{4/3}} + \frac{a_A (A-2Z)^2}{A^2}$$

$$\frac{\partial}{\partial A}\left(\frac{m(A,Z)}{A}\right) = -\frac{Z}{A^2}(m_p + m_e - m_n)c^2 - \frac{1}{3}a_s A^{-1/3-1} - \frac{4}{3}a_c Z(Z-1)A^{-4/3-1}$$

$$+ \frac{a_A}{A^4}\left(A^2 2(A-2Z) - (A-2Z)^2(2A)\right)$$

$$= -\frac{Z}{A^2}(m_p + m_e - m_n)c^2 - \frac{1}{3}a_s A^{-4/3} - \frac{4}{3}a_c Z(Z-1)A^{-7/3}$$

$$+ \frac{2a_A}{A^3}\left(A(A-2Z) - (A-2Z)^2\right)$$

$$A - (A-2Z) = 2Z$$

$$= -\frac{Z}{A^2}(m_p + m_e - m_n)c^2 - \frac{a_s}{3}A^{-4/3} - \frac{4}{3}a_c Z(Z-1)A^{-7/3}$$

$$+ \frac{4a_A}{A^3}(A-2Z)Z$$

$$2 - \frac{7}{3} = \frac{6-7}{3} = -\frac{1}{3} \qquad 2 - \frac{4}{3} = \frac{6-4}{3} = \frac{2}{3}$$

$$A^2 \frac{\partial}{\partial A}\left(\frac{m(A,Z)}{A}\right) = -Z(m_p + m_e - m_n)c^2 - \frac{a_s}{3}A^{2/3} - \frac{4}{3}a_c Z(Z-1)A^{-1/3}$$

$$+ \frac{4a_A}{A}(A-2Z)Z$$

$$-Z(m_p + m_e - m_n)c^2 - \frac{a_s}{3}A^{2/3} - \frac{4}{3}a_c Z(Z-1)A^{-1/3} + \frac{4a_A}{A}(A-2Z)Z$$

$$+ Z(m_p + m_e - m_n)c^2 + a_c \frac{Z}{A^{1/3}}(2Z-1) - \frac{4a_A}{A}Z(A-2Z) = 0$$

$$-\frac{a_s}{3}A^{2/3} - \frac{4}{3}a_c Z(Z-1)A^{-1/3} + a_c Z(2Z-1)A^{-1/3} = 0$$

$$\frac{a_s}{3}A^{2/3} = a_c Z\left((2Z-1) - \frac{4}{3}(Z-1)\right)A^{-1/3}$$

$$2 - \frac{4}{3} = \frac{6+4}{3} = \frac{2}{3}$$

$$\frac{a_s}{3}A = a_c Z\left(\frac{2}{3}Z + \frac{1}{3}\right)$$

$$-1 + \frac{4}{3} = \frac{3+4}{3} = \frac{1}{3}$$

$a_s A = a_c Z(Z-1)$

$Z Z \gg 1 \quad \therefore \quad a_s A = a_c Z(ZZ) = 2a_c Z^2$

$$Z = \sqrt{\frac{a_s A}{2a_c}}$$

$$\frac{\partial}{\partial A} m(A,Z) = m_n c^2 - a_v + \frac{2}{3} a_s A^{-1/3} - \frac{a_c}{3} \frac{Z(Z-1)}{A^{4/3}}$$

$$+ \frac{a_A}{A^2}\left(A\left(2(A-2Z)\right) + (A-2Z)^2\right)$$

$$= m_n c^2 - a_v + \frac{2}{3} a_s A^{-1/3} - \frac{a_c}{3} Z(Z-1) A^{-4/3} + \frac{a_A}{A^2}(A-2Z)(3A-2Z)$$

$$\frac{2}{3} - 1 = \frac{2-3}{3}$$

$$A^{-1/3-1} = A^{-4/3} = -\frac{1}{4/3}$$

$$2A + A - 2Z = 3A - 2Z$$